Brief description of recommended research papers in the “Alzheimer's Dementia Recognition through Spontaneous Speech” editorial

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This research collection addresses the need for non-invasive biomarkers to diagnose and assess Alzheimer's disease by exploring AI-enabled speech and language analysis as a promising diagnostic tool. Most of the 20 papers in the collection focus on two main objectives: classifying individuals with AD versus healthy controls, and predicting cognitive test scores (MMSE). Fourteen papers utilized the ADReSS dataset—a curated, age- and gender-matched subset of DementiaBank's Pitt Corpus—which contains audio recordings and transcriptions of picture descriptions, employing diverse analytical approaches and machine learning models.

Antonsson et al. - This study used a challenging discourse task to examine semantic language production across groups with varying cognitive impairment levels and healthy controls. Combining quantitative oral lexical retrieval tests with qualitative narrative analysis, researchers aimed to distinguish between cognitively stable and cognitively declining individuals. Both assessment methods revealed group differences, though oral lexical retrieval tests proved most effective at differentiating stable from declining populations.

Clarke et al. - This study evaluated computational speech analysis for early Alzheimer's disease and mild cognitive impairment detection using five different connected speech tasks with 25 healthy controls and 25 AD/MCI participants. Researchers extracted linguistic features from picture descriptions, map reading, narrative recall, procedural recall, and wordless picture book narration, then trained machine learning models to classify groups. The findings suggest that over-reliance on particular tasks may limit understanding of how early AD pathology affects speech, highlighting the importance of diverse speech task contexts for improved diagnostic accuracy.

Balagopalan et al. - This study compared two machine learning approaches for automatically detecting Alzheimer's disease from speech: domain knowledge-based methods (extracting clinically relevant linguistic and acoustic features) and transfer learning with pre-trained BERT language models. Using a demographically balanced dataset of 156 older adults (78 with AD, 78 healthy controls) completing a picture description task, both approaches significantly outperformed baseline models. Linguistic features alone achieved comparable or better performance than models combining acoustic and linguistic features. Interpretability analysis of BERT revealed meaningful attention patterns, such as higher attribution to important content units and filler words.

Chlasta and Wołk - This study addressed Alzheimer's dementia detection using the ADReSS Challenge dataset of spontaneous speech with two approaches: VGGish, a pre-trained deep learning model for audio feature extraction paired with Scikit-learn classifiers, achieving 59.1% accuracy and DemCNN, a novel PyTorch convolutional neural network operating on raw waveforms, achieving 63.6% accuracy.

De Looze et al. - This study investigated whether temporal characteristics of speech during a collaborative referencing task with caregivers could serve as biomarkers for MCI and AD. The researchers found that MCI and AD were characterized by slower speech rate and slower turn-taking in conversation, correlating with deficits in episodic memory, semantic/lexical function, executive function, and visuospatial abilities, as well as volume reductions in frontal, temporal, and cerebellar brain regions.

Guo et al. - This study addressed the limitation of small labeled datasets in dementia detection by augmenting DementiaBank's ADReSS challenge data with 1,366 professionally transcribed Cookie Theft Task recordings from the Wisconsin Longitudinal Study (WLS). The findings support the utility of integrating extrinsic normative data from other sources to enhance deep learning model performance for AD classification tasks.

Haulcy and Glass - This study compared multiple machine learning approaches for AD detection and severity assessment using the balanced ADReSS dataset of spontaneous speech. Researchers evaluated five traditional classifiers, convolutional neural networks, and long short-term memory networks trained on audio features and text features. he results demonstrate that spontaneous speech analysis using deep learning-based text representations is effective for both AD classification and predicting neuropsychological assessment scores.

Jonell et al. - This study describes a multimodal system for non-invasive AD screening. Based on preliminary data from 25 patients, the researchers extracted measures from the multimodal recordings and correlated them with clinical assessments and established biomarkers.

Laguarta and Subirana - This paper introduces the Open Voice Brain Model (OVBM), a novel audio processing architecture that improves AD detection from spontaneous speech while simultaneously addressing multiple diseases. Applied to AD diagnosis, OVBM achieved 93.8% accuracy using raw audio alone while generating personalized saliency maps for longitudinal disease tracking across 16 biomarkers.

Lindsay et al. - This study enhanced the clinical validity and generalizability of NLP and machine learning approaches for detecting AD-related language impairment by incorporating multilingual feature analysis. Through inferential statistics, the authors identified language features that were significant in both English and French, representing generalizable markers of cognitive language impairment in AD. The multilingual model distributed errors equally between languages, supporting clinical relevance.

Mahajan and Baths - This study addresses the limitations of current early AD diagnosis methods by developing an end-to-end deep learning approach for AD detection from spontaneous speech. They introduced Speech-GRU, a recurrent neural network architecture that combines acoustic features into a unified vector representation.

Martinc et al. - This study presents a multimodal machine learning approach for Alzheimer's dementia detection from spontaneous speech. The researchers employed Active Data Representation (ADR), a recently developed technique that fuses acoustic and textual features at sentence and word levels, capturing the temporal interplay between paralinguistic (acoustic) and linguistic (textual) components of speech.

Meghanani et al. - This study extends prior work on Alzheimer's dementia detection by shifting focus from acoustic features to linguistic features extracted from spontaneous speech transcriptions. he researchers compared two n-gram-based models. The results demonstrate that n-gram-based linguistic features are effective for AD detection and that fastText models offer a practical advantage: their shallow architecture enables training and evaluation several orders of magnitude faster than deep learning approaches while maintaining competitive or superior performance compared to baseline methods.

Millington and Luz - This study analyzes word co-occurrence networks constructed from spontaneous speech transcripts in the ADReSS challenge dataset to distinguish Alzheimer's disease patients from healthy controls. The researchers examined network structural properties and found that AD patient networks exhibited lower heterogeneity and centralization but higher edge density compared to control networks. Network structure alone achieved 66.7% accuracy for AD classification.

Nasreen et al.- This study investigates conversation analysis features—disfluency patterns and interactional structures—as noninvasive biomarkers for detecting Alzheimer's disease. The findings demonstrate that disfluency and interactional patterns have significant discriminative power for AD detection and show promise for integration into noninvasive clinical assessment tools.

Parvin et al.- This randomized controlled trial examined whether 12 weeks of dual-task training improves cognitive function, physical performance, and brain activity in Alzheimer's disease patients.

Sadeghian et al.- This paper addresses the significant gap between theoretical research on speech-based dementia detection and practical clinical implementation. The authors propose developing an easy-to-use, non-invasive, inexpensive speech-based diagnostic test for dementia that can be administered in clinical offices or home settings. A key contribution is the development of a fully automatic speech recognition system specifically tuned for this application, incorporating automatic punctuation generation to improve transcription accuracy.

Shah et al.- This study develops machine learning models to detect Alzheimer's dementia and estimate cognitive decline using speech samples from the DementiaBank database. The approach explored multiple dimension reduction techniques, training algorithms, and feature fusion strategies to leverage both acoustic and linguistic features extracted from speech.

Yuan et al. - This study leverages fine-tuned Transformer-based language models (BERT and ERNIE) to detect Alzheimer's dementia by modeling pauses, disfluencies, and language irregularities in spontaneous speech.

Zhu et al. - This study systematically evaluates transfer learning approaches for detecting Alzheimer's dementia using ADReSS. Results revealed that text-based transfer learning models significantly outperformed audio-based models, likely because pre-trained language datasets share greater linguistic similarity with the Cookie Theft transcripts than audio pre-training corpora do with dementia speech.